Unit 3 - Deep Learning Basics

Machine Learning | Unit 3

What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses neural networks with many layers (deep neural networks) to learn representations of data with multiple levels of abstraction. It has revolutionized fields like computer vision, natural language processing, and speech recognition.

Convolutional Neural Networks (CNN)

CNNs are specialized neural networks designed for processing grid-like data such as images. They use convolutional layers to automatically learn spatial hierarchies of features, from edges and textures at lower layers to complex objects at higher layers.

CNN Components

Convolutional Layer

Applies filters/kernels to input to create feature maps. Each filter detects specific features like edges, corners, or textures.

Pooling Layer

Reduces spatial dimensions to decrease computation and control overfitting. Max pooling and average pooling are common types.

Fully Connected Layer

Final layers that perform classification based on extracted features. Every neuron connects to every neuron in the next layer.

Recurrent Neural Networks (RNN)

RNNs are designed for sequential data where order matters (text, time series, speech). They have connections that form cycles, allowing information to persist. LSTM (Long Short-Term Memory)

and GRU (Gated Recurrent Unit) are improved variants that solve the vanishing gradient problem.

Transfer Learning

Transfer Learning uses a pre-trained model on a large dataset as a starting point for a new task. Instead of training from scratch, we fine-tune the existing model. Popular pre-trained models include VGG, ResNet, InceptionNet for images and BERT, GPT for text.

Popular Deep Learning Frameworks

- TensorFlow - Google, production-ready, good deployment tools
- PyTorch - Facebook/Meta, research-friendly, dynamic computation graph
- Keras - High-level API, runs on top of TensorFlow
- FastAI - Built on PyTorch, simplifies training